Energy- and Area-efficient Fe-FinFET-based Time-Domain Mixed-Signal Computing In Memory for Edge Machine Learning

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Abstract**—**In this work, for the first time, ferroelectric (FE)-FET is proposed to implement time-domain (TD) computing in memory with nonvolatility (nvCIM) for both multiply-accumulate (MAC) operation and activation function with the record highest area and energy efficiency. Benefiting from the three-terminal transistor structure, the fabricated FeFET based on 14nm-node FinFET can function as both a nonvolatile element for weight storage and a controllable switch for neural network inputs modulation simultaneously, thereby realizing local multiplication of MAC in the proposed FE delay unit with only 2T-1FeFET. The novel dual-edge operation is also demonstrated for TD MAC, enabling further energy efficiency improvement. Furthermore, by utilizing the physics of time-dependent gradual polarization switching, FeFET as the activation function for TD nvCIM with additional memory trace behavior is experimentally demonstrated with only single transistor. Based on the proposed FE-based TD nvCIM design, high-accuracy pattern recognition and accelerated deep reinforcement learning are also demonstrated with scaled voltage of 0.5V, providing a promising area- and energy-efficient mixed-signal neural network for edge AI.

I. Introduction

Computing in memory (CIM) architectures by utilizing non-volatile memory (nvCIM) such as RRAM [1] or PCM [2], which perform multiply-accumulate (MAC) operations in a voltage-domain analog way [1-3], have triggered a lot of attention for artificial neural networks (ANNs). However, relatively high VDD to accommodate the dynamic range of amplitude signals, as well as complex data-conversion analog blocks with large power, are inevitable for voltage-mode nvCIM [1-3]. To address the problem of voltage scalability, time-domain (TD) mixed-signal computing, by utilizing the cumulative propagation delay of cascaded inverter chain to realize the accumulation of MAC, has been regarded as a promising route to significantly improve the energy efficiency, along with high compatibility with digital circuits [4-7]. Nevertheless, only TD-SRAM has been implemented now with the need of extra current-starve circuits [4-6] or variable-capacitance block [7] in the TD cell to realize multiplication of MAC, suffering from high area and energy consumption.

In this work, TD-nvCIM based on ferroelectric FET (FeFET) is proposed and experimentally implemented for the first time. By utilizing the three-terminal structure of the fabricated Fe-FinFET, the local multiplication of MAC can be realized with only 2T-1FeFET, as well as dual edge operation for further energy saving. Additionally, the activation function with additional memory trace behavior is also realized with only one FeFET. Based on the proposed design, high-accuracy pattern recognition and continuous reinforcement learning (RL) are also demonstrated with the record highest area and energy efficiency at 0.5V supply voltage.

II. Design and Experiments of FeFET TD CIM

Fig.2a shows the proposed FeFET based TD-nvCIM array architecture, which performs the accumulation of TD MAC ($y_j = \sum_i x_i \cdot w_i$) in ANN by using cascaded FeFET-based delay units (FE-DUs) of all rows in parallel with considerable signal dynamic range. The FE-DU is composed of an CMOS inverter and a FeFET (Fig. 2b). With large program voltage (V_P), the weight w can be stored in FeFET with different non-volatile polarization states. With a limited gate voltage on FeFET for inputs x modulation (V_X), the FeFET will be operated in non-destructive read mode. When V_X is relatively large, corresponding to the situation of input $x = 0$, the FeFET is turned on with larger drive current than the pull-down NMOS, resulting in small and weight-irrelevant delay time, as the multiplication result of $x \cdot w = 0$. By applying a relatively small V_X of ($x = 1$), the pull-down current and delay time will be determined by weight stored in FeFET, represented as the multiplication result of $x \cdot w = w$. Therefore, multiplication of $x \cdot w$ can be achieved locally by using only one FeFET and be further accumulated through the cascade of FE-DUs. Furthermore, for the temporal information of TD MAC, the FeFET can also be utilized to implement sign activation function due to the time-dependent gradual polarization switching behavior [8] as shown in Fig. 2c.

In this work, the FeFET based TD-nvCIM MAC design is experimentally implemented based on the baseline FinFET in 14-nm technology node. FinFET device with relatively thick gate oxide is designed to satisfy the capacitance matching with $Hf_{0.5}Zr_{0.5}O_2$ FE layer in the gate stack of Fe-FinFET.

III. Fe-FinFET-based Time-domain MAC

A. Characteristics of Fe-FinFET for FE-DU

When applying relatively large gate sweeping voltage on the fabricated Fe-FinFET of the work, the I_d-V_g curves show the typical counterclockwise hysteresis behavior due to the non-volatile polarization switching behavior of HZO (Fig. 3a). In addition, the modulated channel conductance of FeFET can be retained after program/erase with negative/positive pulses applied (Fig. 3b), indicating the Fe-FinFET as the non-volatile storage element for weight w in FE-DU.

After programming, the measured I_d-V_g curves and I_d-V_d curves of the fabricated Fe-FinFET under the relatively small gate voltage range (Fig. 4) show non-destructive behavior with negligible hysteresis due to the partial polarization switching of FE and large I_{ON}/I_{OFF} ratio for variable input x modulation. Moreover, the threshold voltage (V_{TH}) of device can be modulated for multilevel weight storage under different V_P.

B. Fe-FinFET based stage for local TD multiplication

Based on the fabricated Fe-FinFET, the FE-DU stage is further experimentally constructed for local TD multiplication with multilevel weight states under scaled VDD.

FE-DU based local TD multiplication: In the initial phase, the weight is programmed in Fe-FinFET in the TD nvCIM with different V_P applied (Fig. 5ac), and the weight readout results indicate the programmable propagation delay time of FE-DU (Fig. 5bd). The input x of ANN is represented by V_X. As shown in Fig. 6, with $V_{X=0}$, the weighted Fe-FinFET is fully turned on with large drive current, and thus the delay time of FE-DU will not be affected by the weight stored in Fe-FinFET. For $V_{x=1}$, the drive current of Fe-FinFET is determined by the stored weight. For the fabricated FE-DU in multiplication phase (Fig. 7), since the Fe-FinFET determines the pull-down current and the propagation delay when $V_{X=1}$=0.7V, the rising edge of FE-DU can be modulated with different propagation delays corresponding to different weights, while the delays are unregulated by weight and is very small originated from the intrinsic delay of the inverter when $V_{X=0}$=1.3V. Therefore, the FE-DU can realizes the local multiplication $(x \cdot w)$.

Moreover, the inverter of each FE-DU can also be reused by several weighted Fe-FinFETs from different layers in ANN, which is beneficial for further area saving. By applying unselected voltage (V_g=0V) to turn off the unselected weighted Fe-FinFETs, the rising edge of computing pulse cannot be effectively reversed as the unselected FE-DU (Fig. 8).

Multilevel weight representation of FE-DU: With multiple voltage pulses applied to the gate of Fe-FinFET, the FE polarization can be gradually switched in succession, resulting in the continuous modulation of V_{TH} of FeFET (Fig. 9ab). Therefore, when applying V_p pulse trains to Fe-FinFET of FE-DU stage and measuring the propagation delay, it is shown that the weighted delay can be modulated gradually (Fig. 9cd), indicating multilevel weight storage with high density.

Moreover, with scaled VDD, the consistent propagation process of FE-DU is demonstrated (Fig. 10), and the voltage transfer characteristics (VTC) of the inverter based on FinFET also show superior gain, indicating the great potential of the FE-TD nvCIM for ultralow power and high energy efficiency.

C. Dual-Edge operation and cascade FE-DUs for MAC

Based on the proposed FE-DU, a novel dual edge operation of cascaded FE-DUs is further experimentally demonstrated.

Dual-Edge operation of FE-DU: In conventional SRAM based TD computing [5-6], only single transition of computing pulse is used for MAC, and an additional inverter in each stage is needed to reverse the computing pulse polarity for next stage (Fig. 1b). In our work of FE-DU, both dual transitions are used for MAC (Fig. 11b). As shown in the measured results of FE-DU (Fig. 11c), for one stage in the delay chain, the rising edge is delayed by the multiplication result in Fe-FinFET and reversed to the falling edge, which can be reversed back again by the PMOS of following stage. Therefore, the MAC value is the sum of time delay (T_{MAC}) of front and back edge respectively caused by odd stages and even stages in the delay chain, and dual transitions of computing pulse are utilized in the Fe-FinFET based TD CIM array for further energy saving.

Cascade of the FE-DUs for TD MAC: In our experiments, MAC of four stages are further implemented and demonstrated with varied input x and weight w. Two types of weight are set in weight program phase. The measured MAC value is the sum of delay time of front edge caused by odd stages and back edge caused by even stages (Fig. 12), indicating the intrinsic MAC process in the FE-DU chain. For more cascade stages, to further compensate the intrinsic inverter delay, a row of delay reference is also designed with the same amount of stages (Fig. 2a), and the more accurate MAC value is the delay difference between the output edges of computing line and reference line.

IV. FeFET-based Time-domain Activation

In the proposed TD CIM of this work, the MAC results of temporal information are also designed to be processed by a single FeFET device for nonlinear activation function.

The voltage pulse train is applied to the gate of FeFET with partial forward polarization during T_{MAC}. Only when the T_{AMC} is large enough, can the input pulses make the FeFET device channel sufficiently conductive and thus an activation triggers output, realizing the sign activation (Fig. 13). In addition, the initial base state of the activation FeFET can be modulated and retained, which can further adjust the threshold bias of TD activation (Fig. 14). Therefore, the TD activation is realized by only one FeFET. Moreover, in FeFET, since the accumulated partial polarization can be back-switched by depolarization field [9-10], a memory trace process can be realized as shown in Fig. 15, resulting in the faster activation with immediate input. It indicates that the reinforcement learning ability of edge agents can be enhanced in our design without the need of additionally large area capacitor in conventional method [11].

V. FeFET based TD nvCIM Macro

Based on the proposed FeFET-based TD nvCIM, the ANN based pattern recognition with weight quantization of FE-DU is demonstrated with high accuracy comparable with floating point weight (Fig. 16). Due to the memory trace process of FeFET based activation, the deep reinforcement learning is enhanced with convergence rate without extra hardware cost (Fig. 17). The proposed FeFET based TD nvCIM MAC array of 128×100 is simulated based on our developed Fe-FinFET model, showing that the MAC function can be realized with multilevel weights of Fe-FinFET and large dynamic range of accumulation results in TD (Fig. 18-19). In addition, the energy efficiency of one TD-MAC delay chain is also evaluated and compared, showing significantly improvement with VDD scaling and fin-number decreasing (Fig. 20). For MAC operation of this work, the energy efficiency is the record highest among all the TD computing as 51318TOPS/W.

VI. Summary

This work reports the first experimental demonstration of FeFET-based area- and energy-efficient time-domain CIM with 3T FE delay unit for MAC and one FeFET for activation funtion, enabling dual-edge MAC operation with multilevel weight and time-domain activation with memory trace function for edge deep learning respectively (Table 1). Based on the proposed design, high-accuracy pattern recognition and accelerated deep RL are also demonstrated under scaled VDD of 0.5V with the record highest energy efficiency, showing its great potential for mixed-signal neural network for edge AI.

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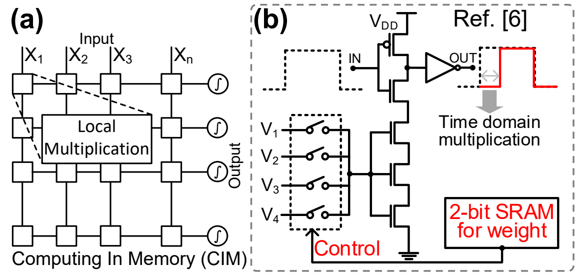

(C)	Conventional methods		This work
Signal domain	Digital	Analog (Voltage)	Analogy (Time)
Digital compatibility	Yes	No	Yes
Technology/Voltage scaling	Yes	Hard	Yes
Energy efficiency	Low	Relatively high	High
Activation Function	Digital cell	Data converter	One FeFET

Fig. 1 (a) Computing in memory (CIM) architecture with local multiplication. (b) The SRAM based time-domain (TD) computing stage [6]. (c) Advantages of TD mixed-signal computing in this work compared with other types of computing.

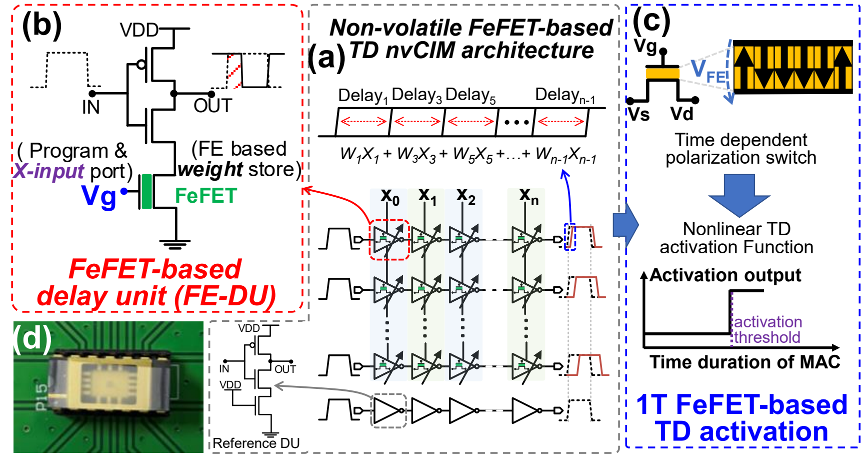

Fig. 2 (a) The proposed nonvolatile FeFET-based TD nvCIM architecture with FeFET-based delay unit (FE-DU) array and a reference delay chain. (b) The topology of FE-DU with a FeFET and an inverter for TD computing. (c) The principle of 1FeFET based TD-activation. (d) The hardware implementation of Fe-FinFET based TD circuit.

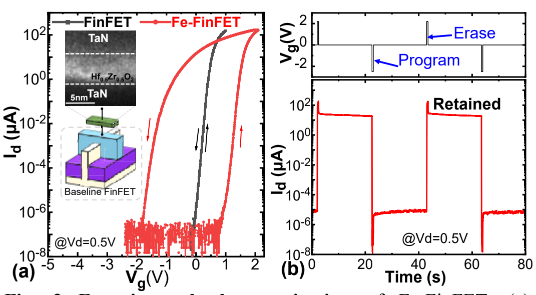

Fig. 3 Experimental characterization of Fe-FinFET. (a) Measured I_d-V_g curve of Fe-FinFET with relatively large gate sweeping voltage range, and insert is TEM of FE layer. (b) The channel conductance modulation with nonvolatility.

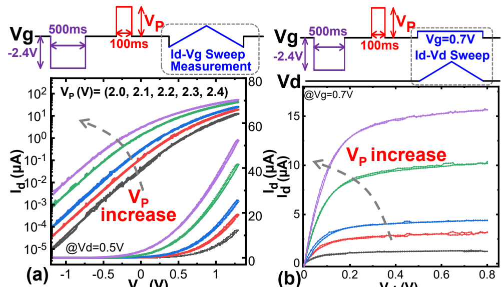

Fig. 4 Measured threshold voltage (V_{TH}) modulation of Fe-FinFET. (a) I_d-V_g curves and (b) I_d-V_d curves with slow forward-and-back sweeping of relatively small gate voltage range, showing non-destructive behavior.

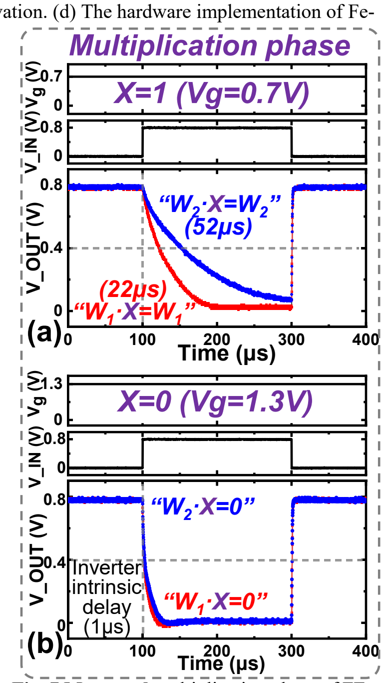

Fig. 7 Measured multiplication phase of FE-DU with weight stored in Fe-FinFET and input x applied to V_g of Fe-FinFET. The time delay is modulated by multiplication between weight with (a) x=1 and (b) x=0.

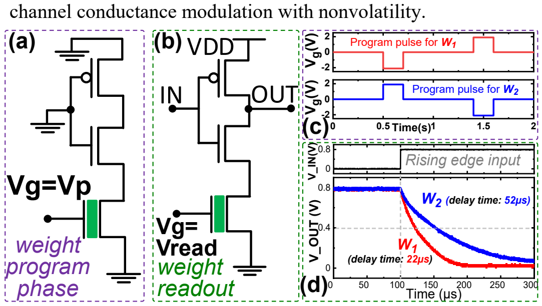

Fig. 5 (a) The weight program phase with (c) different V_P applied. (b)(d) **Measured** readout propagation delay process of FE-DU with oscilloscope. The delay determined by weight is the time for voltage pull-down to VDD/2. (V_{read}=0.7V).

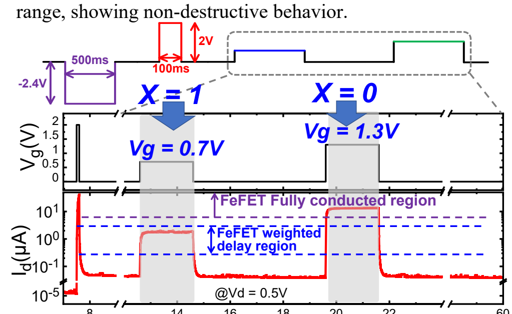

Fig. 6 Measured operation mode of Fe-FinFET in FE-DU with varied input x value of ANN without affecting the storage state. ($V_{x=0}$=1.3V, $V_{x=1}$=0.7V in this work).

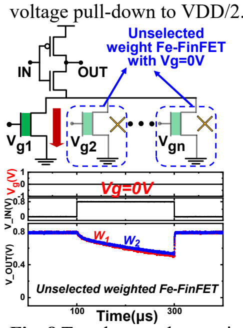

Fig. 8 Topology and operation principle of sub-array design based on FE-DU.

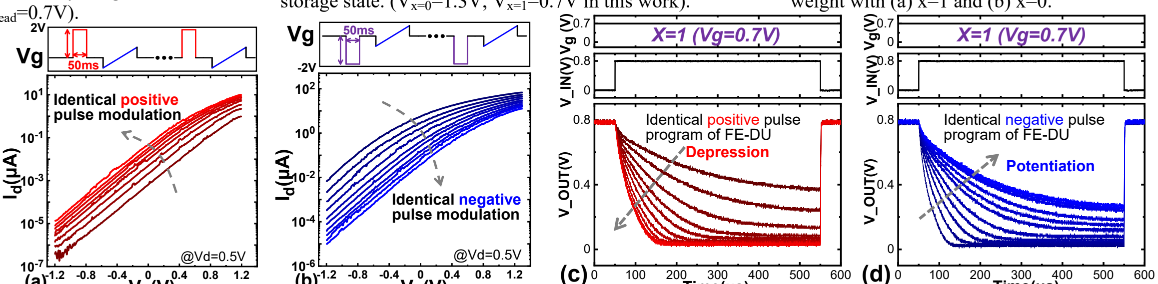

Fig. 9 Measured multilevel weight representation of Fe-FinFET and FE-DU. With multiple (a) positive or (b) negative pulses applied to the gate of Fe-FinFET, the I_d-V_g curves is measured under relatively small gate voltage range, showing the (a) decrease or (b)increase of threshold voltage (V_{TH}). With (c) positive or (d) negativeVP pulses applied to Fe-FinFET of FE-DU in weight program phase and measuring the propagation delay with V_g=V_{read}, it shows (c) depression or (d) potentiation process of multilevel weight.

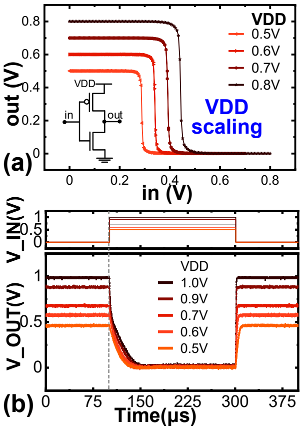

Fig. 10 Measured (a) voltage transfer characteristic (VTC) of FinFET-based inverter, and (b) the propagation process of FE-DU with VDD scaling.

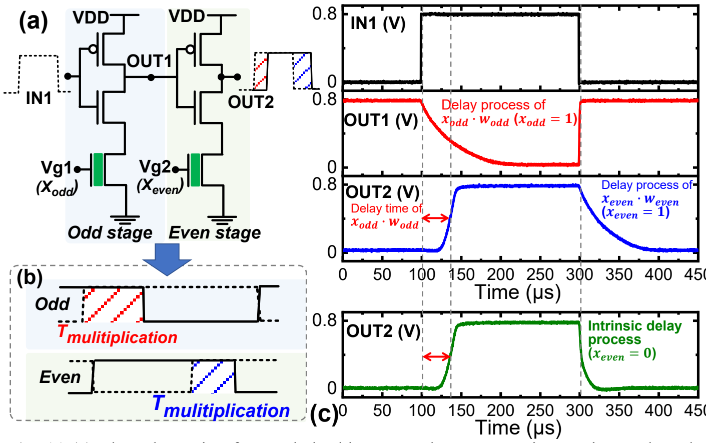

Fig. 11 (a) The schematic of cascaded odd stage and even stage in Fe-FinFET based TD CIM array. (b) The schematic of time domain modulation of odd/even stage. (c) **Measured** dual-edge operation of FE-DU with programmed weight. The delay of front /back edge is caused by multiplication in odd/even stage.

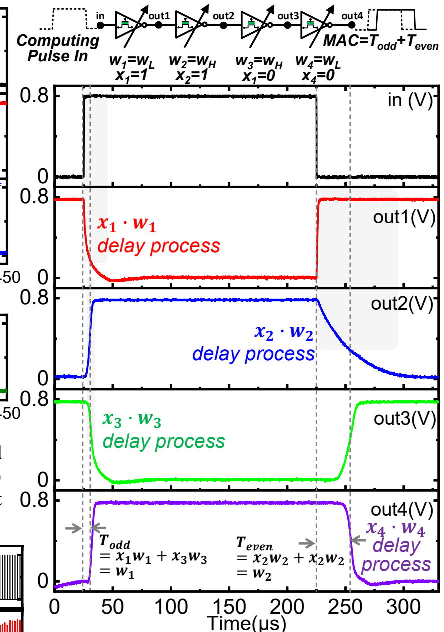

Fig. 12 Measured cascade of the FE-DUs for TD MAC. W_L/W_H is weight of low/high delay modulation, which is set in weight program phase. The propagation process shows local multiplication and intrinsic accumulation of MAC in FE-DU chain with dual-edge operation.

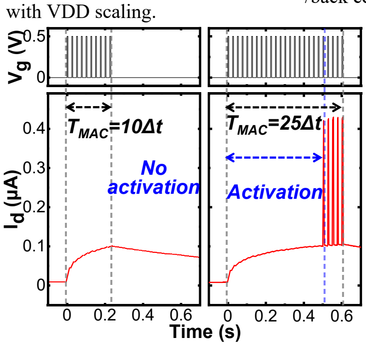

Fig. 13 Measured temporal MAC information process with FeFET-based activation, showing sign activation function. The pulse voltage applied to FeFET is 0.5V.

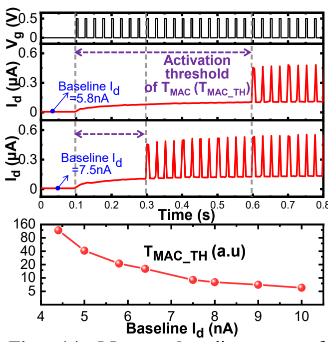

Fig. 14 Measured adjustment of activation threshold by modulating initial base polarization of FeFET, reflected as baseline I_d at rest state.

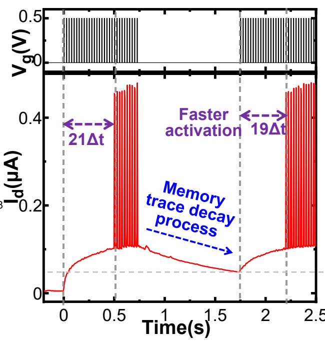

Fig. 15 Measured memory trace decay process of FeFET-based activation, showing enhancement to immediate activation process.

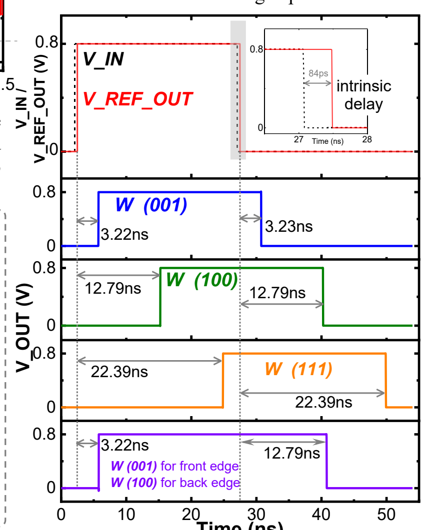

Fig. 18 The signal timing diagram of chains in 128×100 FE-DU array. The intrinsic delay can be compensated by taking REF_OUT as reference. When weight of a line are all programed to (001)/(100)/(111) with input of each stage set as x=1, the MAC results are correctly reflected in the time delay, and dual-edge operation is realized.

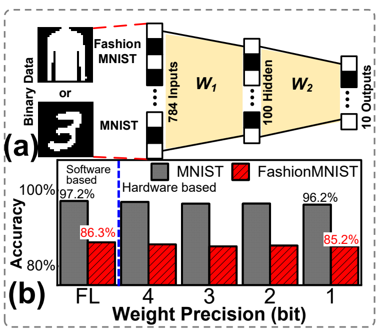

Fig. 16 (a) The architecture of ANN based on proposed FE-TD CIM with multilevel FE-DUs and FeFET-based sign activation function. (b) The pattern recognition based on proposed hardware shows high accuracy close to baseline of software.

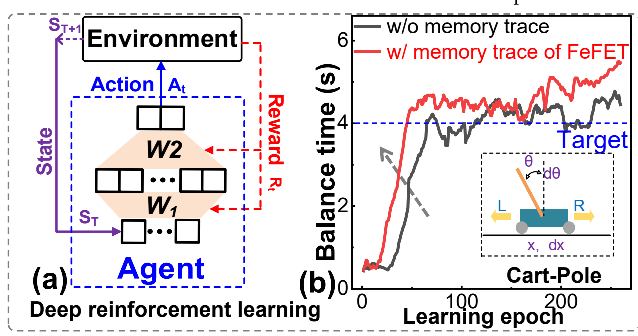

Fig. 17 (a) The deep reinforcement learning frame based on proposed FE based TD MAC and FeFET-based activation with memory trace for agent of deep neural network. The real-time weight update dependent on reward can be realized due to decoupled program and read path of three-terminal Fe-FinFET of FE-DU. (b) The learning process of cart-pole task is accelerated with memory trace (eligibility trace) process of proposed FeFET-activation.

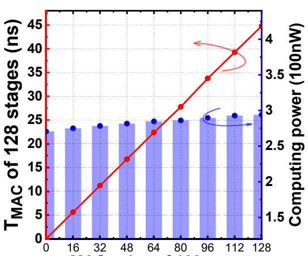

Fig. 19 The FeFET-based TD MAC shows good linearity and signal dynamic range, and the power is independent with data.

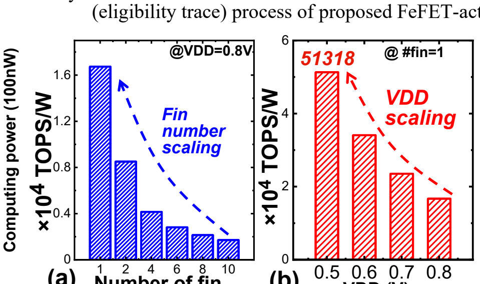

Fig. 20 Energy-efficiency TOPS/W (1bit MAC=2OPs) of proposed FeFET based TD nvCIM with varied device fin numbers and VDDs, which according with dynamic energy consumption $E = CV^2$.

Table. 1 Benchmark of the proposed FE-based TD CIM with other analogy computing methods

	Ref. [3]	Ref. [6]	Ref. [5]	Ref. [4]	**This work**
Analog Domain	Voltage	Time			**Time**
Weight memory cell	FeFET	8T-SRAM	SRAM	6T-SRAM	**Fe-FinFET**
NVM	Yes	No			**Yes**
Cell size	1T+1R	＞19T	＞55T	12T	**3T (1FeFinFET+2T)**
Cell size/bit	(1T+1R)/bit	＞9T/bit	＞11T/bit	12T/bit	**<1T/bit**
Activation	ADC	Digital logic blocks			**One FeFET**
DNN Application	--	ImageNet	MNIST	MNIST	**Image recognition & Reinforcement learning**
MAC Energy efficiency (TOPS/W)	13714	119.7	48.5	716	**51318 TOPS/W**